

# INTEGRATED CLOUD SYSTEMS + CYBERSECURITY

The Integrated Cloud, Systems and Cybersecurity Track prepares students to deploy, manage, automate, secure and defend modern enterprise and cloud environments. It combines system administration, networking, database administration, cloud architecture, DevOps and DevSecOps, infrastructure as code, virtualization, containers, security operations, penetration testing, incident response, digital forensics, GRC and AI-assisted automation.

## JOB PROFILES

**Cloud Administrator - System Administrator - Network Engineer - DevOps Engineer - Database Administrator - Data Center Engineer - Cybersecurity Engineer - Cyber Threat Analyst - Penetration Tester - Incident Responder - Digital Forensics Analyst - GRC**

### TOTAL HOURS

**1,620 hrs**

Proposed integrated plan

### DELIVERY

**Blended**

Online & on-site

### UNIVERSITY GRADE

**Good**

Minimum grade required

### CERTIFICATIONS

**7**

Pathway options

## + Where graduates work



## + Certifications earned

- A RHCSA** - Red Hat Certified System Administrator
- A CKA** - Certified Kubernetes Administrator
- A AWS-SAA-C03** - AWS Certified Solutions Architect - Associate
- A NCP-MCI** - Nutanix Certified Professional - Multicloud Infrastructure
- A HCCDA** - Huawei Certified Cloud Developer Associate
- A eWPT / eMAPT** - Web & Mobile Application Penetration Testing
- A eCIR / NSE 4** - Incident Response and Network Security

## + Selection process

- 01 IQ & English Exam**  
Online exam covering problem-solving aptitude and English proficiency.
- 02 Technical Exam**  
Computer-based assessment covering the program core technical fundamentals.
- 03 Technical Interview**  
Interview with the technical committee.
- 04 Interpersonal Interview**  
Candidates who pass the technical interview are invited to this final stage.

## + Who may apply

Open to graduates in Computer Engineering, Communications Engineering and Computer Science with a minimum grade of Good or higher.

## INTEGRATED CLOUD + SYSTEMS + CYBERSECURITY

PROFESSIONAL TRAINING PROGRAM

**+ Program content - 1,620 hours**

| CATEGORY              | HOURS   | SHARE | KEY COURSES / AREAS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|---------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ● Fundamental Courses | 300 hrs | 19%   | OS, Linux and Bash; Python, OOP and scripting; database fundamentals and MySQL; client-side technologies; networking fundamentals; cloud computing and virtualization foundations; cybersecurity fundamentals; Generative AI and prompt engineering.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ● Core Courses        | 963 hrs | 59%   | Enterprise networking and routing/switching; Windows Infrastructure and Active Directory; Red Hat administration I-III, Ansible and high availability; web server administration; VMware vSphere/NSX, OpenStack, Nutanix, Azure, GCP and AWS SysOps; storage, data center design and Veeam backup; Docker, Kubernetes and OpenShift; Git, Terraform, IaC, DevOps, Advanced DevOps and DevSecOps; Oracle SQL/PLSQL and DBA; cloud security; firewalls, Metasploit, ethical hacking and bug hunting; web/mobile penetration testing; malware analysis and reverse engineering; OT, application and AI security; GRC, SOC fundamentals, threat hunting, forensics, Active Directory pentesting and incident handling. |
| ● Soft Skills         | 57 hrs  | 4%    | Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills and Professional Demeanor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ● Graduation Project  | 300 hrs | 19%   | Integrated capstone: build a secure enterprise/cloud platform with networking, identity, IaC, CI/CD, containers, monitoring, backup and recovery, hardening, vulnerability assessment, incident response playbooks and governance documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

**+ Student Deliverable**

Every graduate must obtain at least one international certification relevant to their selected pathway - RHCSA, CKA, AWS-SAA-C03, NCP-MCI, HCCDA, eWPT/eMAPT, eCIR or NSE 4 - confirming professional readiness.

**+ Before you apply**

Explore these resources before your interview. Applicants are expected to have

- > **Introduction to Computer Networks**  
<https://maharatech.gov.eg/course/view.php?id=37>
- > **Linux Essentials**  
<https://maharatech.gov.eg/course/view.php?id=2155>
- > **VMware Foundation**  
<https://maharatech.gov.eg>
- > **Introduction to Cloud Computing**  
<https://maharatech.gov.eg>
- > **Introduction to Cybersecurity**  
<https://maharatech.gov.eg>
- > **ITI Values & Mission**  
<https://iti.gov.eg/about-us>

APPLY AT

**ITI.GOV.EG**

Applications open each intake period. Check website for dates

Note: Proposed integrated structure synthesized from the three uploaded ITI track sheets; hours are redistributed because source PDFs do not provide course-level hours.